

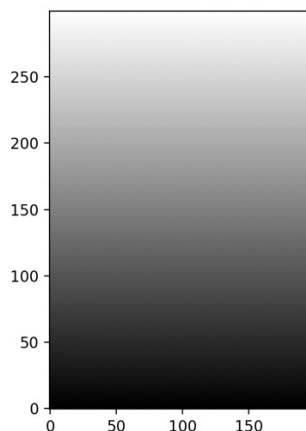
UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Practicum 1: Python practice
September 11, 2026

Welcome to your first practicum exercises!

We will start by doing three exercises that you must hand in at the end of class.

1. **(10 points)** Start by coding this yourself. Don't use AI yet.

- Make a 300×200 Float64 array. Each array element should contain its own x coordinate, starting with $x = 0$ for column 0 and ending with $x = 299$ for column 299 (i.e., column 0 has 200 zeros).
- How do you do that with a loop?
- How do you do that without any loops?
- Display the array using the matplotlib function for showing the image with a cmap of 'gray'.
- Examine several randomly placed array elements with Python to be sure the values match the y coordinate in the print statements. Be careful that the final array has the correct data type.
- Make sure that the lower left corner of your plot is 0,0! Your image should look like this:



Take a screen shot showing both your Python code and plot windows.

2. **(10 points)** Now repeat the above using the help of AI. Write the prompts in your log for today's class and copy-paste to the AI prompt. How do you need to write your prompts to get the wanted outcome? Take a screenshot of your prompt(s), the AI-generated code, and its plots.

3. **(5 points)** If you haven't done it yet, install astropy
<https://docs.astropy.org/en/stable/install.html>

Download and read FITS file `m42_40min_ir.zip` from the `practicum/` folder. Don't worry, the FITS function knows how to read the zip file and you don't need to unzip it yourself (see also: <https://docs.astropy.org/en/stable/io/fits/>). This file comes from the NOIRLab FITS Liberator project. Plot it using the appropriate matplotlib function using a grayscale colormap. Make sure that the lower left corner of your plot is 0,0! What object is it?

- Annotate the image by placing the object's name and your name as a title. Make sure the title fits within the image boundaries!
- Add appropriate axis labels.
- Use the appropriate python command to save the final plot in a PNG format. Remember to use an appropriate name following the class's conventions.